

Research Paper Writing Seminar I (RPW1)

Session 01: Introduction to RPW1

AY 2026, S2 (Q3 and Q4)

RPW1 Team

University of Aizu

Agenda

- Course overview
- Research papers and why we write them
- The research and writing process
- Peer review and revision
- Q&A and recap

Our university motto

to Advance Knowledge for Humanity

Our founding principle calls us to **invent** and **discover** for the peace and prosperity of humankind.

My personal take

- We are here to invent new technologies, discover useful things, and create real value for our local community and the world.
- We are NOT here simply to publish as many papers as possible.
- **Papers are a way to communicate our discoveries — not the ultimate goal of our research.**

Teaching team

The AY2026 teaching team includes:

- KACHI Yasuyuki: Mathematics
- ROY Debopriyo: Linguistics
- VIGLIETTA Giovanni: Theory of Informatics
- NASSANI Alaeddin: Augmented Reality
- DANG Nam Khanh: Computer Architecture

We also have several invited lecturers!

RPW1 Survey



These results describe the 2025 class.

RPW1 Survey: goals and support



These results describe the 2025 class.

Common writing difficulties

- Explaining an idea clearly
- Organizing sections and connecting ideas
- Academic English and word choice
- Choosing a research topic
- Reviewing related work
- Using LaTeX

Course goals

What RPW1 covers

- Understand the structure of a research paper.
- Develop skills for research paper writing.
- Learn citation, submission, and ethical writing practices.

Course scope

- Your supervisor guides your research.
- RPW1 focuses on scientific writing. General English study is separate.

Research fields

The course uses examples from computer science and engineering.

- Computer systems and networks
- Security, software, and databases
- Theory and algorithms
- AI, graphics, and multimedia
- Human–computer interaction
- Other areas of computer science and engineering

Course plan: Sessions 01–06



Check ELMS for the confirmed order and dates. Topics and dates may change.

Course plan: Sessions 07–14



Check ELMS for the confirmed order and dates. Topics and dates may change.

Course information in ELMS

Use the **AY2026 RPW1 ELMS course page** for:

- Class dates, rooms, and schedule changes
- Attendance and grading rules
- Plagiarism and generative AI rules
- Slides, questionnaires, and assignments

Planned class time: periods 9–10, **17:00–18:40**, on Monday, Tuesday, or Wednesday.

Check ELMS and your email for the date of each class and any changes.

Midterm and final assignment



Your final draft must address concerns from all three midterm reviewers.

Individual work and course rules

The AY2026 rules are posted on ELMS:

- Individual authorship and permitted help
- Citation and reuse of research material
- Generative AI use and disclosure
- Submission format and review responsibilities

Ask the instructors if a rule is unclear.

Grading

- **50%: Class activities**, including your peer reviews
- **50%: Paper draft**, with midterm and final rounds

Check ELMS for the detailed rubric and deadlines.

Session 01: Q&A and survey



- Open ELMS: <https://elms.u-aizu.ac.jp/>
- Please complete the questionnaire in 10 minutes.
- The survey closes at **17:50**.

What is a research paper?

Common types of research papers



What a research paper explains



- What problem the research addresses
- How the work relates to earlier studies
- What the authors did
- What evidence supports the conclusions
- What limits remain

Example: A systems paper

Pollux: Co-adaptive Cluster Scheduling for Goodput-Optimized Deep Learning

The first page includes a title, authors, abstract, and introduction.

How does the abstract explain the problem and result?

Example: A machine learning paper

Attention Is All You Need

Look at the title, author list, abstract, and section headings.

How do they help you find the main idea?

Why RPW1?

Why research writing matters



Why RPW1 in the era of LLMs?

- Large language models (LLMs) can help draft and revise text.



- **You need to build and check a clear research argument.**

Benefits of research writing

Academic:

- Develops analytical thinking and study design skills.
- Deepens understanding of the research topic.
- Improves how you explain methods and results.

Professional:

- Helps you communicate your expertise.
- Supports preparation for academic and industrial R&D roles.

Personal:

- Encourages intellectual curiosity.

Skills you can use beyond RPW1

- Write a research paper
- Explain your work in a job interview
- Describe an invention in a patent application
- Write a funding proposal
- Pitch an idea for a startup

Example: LLM training costs

Manager: "Training our large language model costs too much."

Possible replies:

- "Buy cheaper GPUs."
- "Rent cheaper computing resources."
- "Measure the costs first."
- "Reduce our salaries!"

Which reply helps us understand the problem?

Defining the research problem



The problem must be **clearly defined**.

Evidence from earlier work



A trusted source still needs **careful reading**.

A focused research question



Example research question:

- **Can GPU clock settings reduce training energy within a time limit?**

A method and a fair test



Reporting the result



The research and writing process

The research and writing process



A useful research question

A useful question is:

- Clear enough for readers to understand
- Narrow enough for the planned study
- Possible to answer with available resources
- Important to the research community

Example:

- Weak: How does AI improve healthcare?
- Better: How does image augmentation affect a fixed model's COVID-19 classification accuracy on a held-out chest X-ray test set?

Literature review



- Explain how earlier work relates to your question.
- Identify existing research gaps.
- Support the development of your methodology.

We will cover this in Session 02.

Finding relevant papers



[Google Scholar](#) · [IEEE Xplore](#) · [ACM DL](#) · [arXiv](#)

AI search tools can help. Verify papers and references against the original sources.

We will cover this in Session 02.

Paper structure



Section names and order vary by field and venue.

Paper structure (continued)



Some papers combine sections or use different names.

Citations and plagiarism



- Cite your own previous work when you reuse or extend it.
- Be cautious of AI hallucinations, and verify AI-generated claims and references.
- Check permissions before reproducing figures, tables, or text.

Writing tools and generative AI



You remain responsible for the work you submit.

Revision



Ask a reader to explain your main point back to you.

Common problems in a draft

- A broad problem with no clear research question
- A contribution that readers cannot identify
- Claims without enough evidence
- A list of papers without comparison or analysis
- Figures or explanations that are hard to follow
- Missing citations or incorrect formatting

Fix the main argument before polishing individual sentences.

Writing and reference tools

Task	Tools
Writing and formatting	Word, LaTeX, Overleaf
Managing references	Zotero, JabRef, Mendeley
Drafting and revision support	LLMs, where permitted
Similarity checking	Turnitin, iThenticate
Grammar check	Grammarly

Choose tools that fit your supervisor's workflow and the venue.
Check the output yourself.

How will your papers be reviewed?

Peer review

Peer review asks other researchers to assess a paper's quality and validity.

1. Initial screening: editorial office and editor / chair

- The office checks format and required information.
- The editor or chair checks scope and suitability for review.
- The paper may be returned for corrections or **rejected before review**.

2. Handling editor / area or track chair

- Assigns reviewers and manages the review process.
- Considers reports and recommends or makes a decision.

Peer review (continued)

3. Reviewers

- Are invited to evaluate the paper.
- Assess its novelty, technical quality, clarity, and significance.
- Are usually anonymous, depending on the venue's review policy.

4. Decision authority: editor or program committee leadership

- Considers the paper, review reports, and recommendations.
- Makes or approves the **final decision**, under the venue's policy.

Roles and decision authority vary by venue.

Before submission



Submission and review



The number of reviewers and steps depend on the venue.

Anonymous review and conflicts



Possible decisions



A request for revision or an author response does not guarantee acceptance.

Responding to reviews



Support disagreements with reasons and evidence.

A useful review



Be specific, fair, and constructive. Keep reviews confidential.

Reviewers and decision makers

- Reviewers are usually experts in the specific research field.
- Editors and program committee chairs
 - Have broad knowledge of the research area.
 - May not be experts in every specific topic.
 - Usually give significant consideration to the reviewers' recommendations when making the final decision.

Revision and rebuttal: we will cover this in Session 11.

Peer review: overall flow



Typical journal process. Conference review may have fewer revision rounds.

Some modest suggestions

Clear scientific writing



- Precision matters more than impressive language.
- Clear explanations help busy editors and reviewers assess your work.
- Readers need to understand the contribution and its supporting evidence.

A precise sentence



Explain the workload, measurement method, and variation.

Writing for your readers



Conferences and journals



Read the current venue instructions.

Choosing a publication venue

Discuss the choice with your supervisor.

- Does the venue publish work on your topic?
- Who is the intended audience?
- Are the review process and publisher clear?
- Can you meet the format and deadline?
- Are publication costs and policies acceptable?

Use rankings as one source of information.

Read recent papers and assess their relevance.

Learning from published papers



Learn from the structure. Write your own explanation.

Venue rankings and citation metrics

- Computing venue lists: [ICORE \(formerly CORE\)](#) and CCF
- Journal indicators: SCImago Journal Rank and Journal Impact Factor
- Publication citation metrics: [Google Scholar Metrics](#)

These indicators provide context. They do not establish the quality of an individual paper.

Read the papers and discuss venue choices with your supervisor.

Q&A time

Questions and review



What would you like to ask about the course?

Four points to remember

- Be precise.
- Use simple language.
- Improve the paper through revision.
- Focus on quality rather than quantity.

to Advance Knowledge for Humanity

Our founding principle calls us to **invent**
and **discover** for the peace and
prosperity of humankind.

Thank you

Questions?

Research Paper Writing Seminar I (RPW1)

Session 01, AY2026

Next topic: Literature survey and review